

Q1: (a): (5 points)

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{nk} \quad (k=0, 1, 2, \dots, N-1)$$

$$\tilde{X}_3[k] = \sum_{n=0}^{3N-1} \tilde{x}[n] W_{3N}^{nk} \quad (k=0, 1, 2, \dots, 3N-1)$$

$$= \sum_{n=0}^{N-1} \tilde{x}[n] W_{3N}^{nk} + \sum_{n=N}^{2N-1} \tilde{x}[n] W_{3N}^{nk} + \sum_{n=2N}^{3N-1} \tilde{x}[n] W_{3N}^{nk}$$

$$= \sum_{n=0}^{N-1} \tilde{x}[n] (W_{3N}^{nk} + W_{3N}^{(n+N)k} + W_{3N}^{(n+2N)k})$$

$$= \sum_{n=0}^{N-1} \tilde{x}[n] W_{3N}^{kn} (1 + W_{3N}^{kn} + W_{3N}^{2kn})$$

$$= \frac{1}{N} (1 + W_{3N}^{kn} + W_{3N}^{2kn}) \sum_{n=0}^{N-1} W_{3N}^{kn} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{kn}$$

1° When $k = 3m+1$ or $3m+2$ ($m \in \mathbb{Z}$)

$$1 + W_{3N}^{kn} + W_{3N}^{2kn} = 0, \quad \therefore \tilde{X}_3[k] = 0$$

2° When $k = 3m$ ($m \in \mathbb{Z}$)

$$1 + W_{3N}^{kn} + W_{3N}^{2kn} = 3 \quad \therefore \tilde{X}_3[k] = \frac{3}{N} \cdot N \cdot \tilde{X}\left[\frac{k}{3}\right] = 3 \tilde{X}\left[\frac{k}{3}\right]$$

$$\Rightarrow \tilde{X}_3[k] = \begin{cases} 0 & k = 3m+1 \text{ or } 3m+2 \quad (m \in \mathbb{Z}) \\ 3 \tilde{X}\left[\frac{k}{3}\right] & k = 3m \end{cases}$$

(b): (5 points)

By the results of (a) and Figure info, we can get:

$$\begin{cases} \tilde{X}[0] = 3 & \text{and } \tilde{X}[k] \text{ is periodic with } T=2 \\ \tilde{X}[1] = -1 \end{cases}$$

$$\tilde{X}_3[k] = \begin{cases} 9 & k=0 \\ 0 & k=1 \\ 0 & k=2 \\ -3 & k=3 \\ 0 & k=4 \\ 0 & k=5 \end{cases} \quad \text{and } \tilde{X}_3[k] \text{ is periodic with } T=6$$

Then the expression in (a) is verified.

Q2. (a) : (4 points)

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \sum_{n=0}^{\infty} (\alpha \cdot e^{-j\omega})^n$$

$$= \frac{1}{1 - \alpha e^{-j\omega}} \quad (|\alpha| < 1 \text{ otherwise not converge})$$

(b) : (4 points)

$$\tilde{x}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn}$$

$$= \sum_{n=0}^{N-1} \sum_{r=-\infty}^{\infty} x[n+rN] W_N^{kn}$$

$$= \sum_{n=0}^{N-1} \sum_{r=0}^{\infty} \alpha^{n+rN} W_N^{kn}$$

$$= \frac{1}{1 - \alpha^N} \sum_{n=0}^{N-1} (\alpha \cdot e^{-\frac{2j\pi k}{N}})^n$$

$$= \frac{1}{1 - \alpha^N} \cdot \frac{1 - \alpha^N e^{-j2\pi k}}{1 - \alpha e^{-j\frac{2\pi k}{N}}}$$

$$= \frac{1}{1 - \alpha e^{-j\frac{2\pi k}{N}}}$$

(c) $\tilde{x}[k]$ is the ~~result~~ sampling result of $X(e^{j\omega})$

and $\omega_s = \frac{2\pi}{N}$

(2 points)